



Session 2 :Pandas & Numpy for Data engineering

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The Data Chaos Begins...

Among the many tools and libraries available to data engineers, **Pandas** and **NumPy** stand out as foundational libraries in Python that enable seamless data manipulation and analysis. Whether you're building a data pipeline, cleaning raw data, or conducting exploratory data analysis (EDA), mastering Pandas and NumPy is key.



Pandas : The Data Therapist





Common use cases of Pandas in Data Engineering:

- Reading and writing data from various file formats (CSV, Excel, JSON, SQL databases)
- Data cleaning and preprocessing (handling missing data, duplicates, or outliers)
- Transforming data (reshaping, merging, and filtering)
- Summarizing and aggregating data for analytics

Numpy: The Math Beast



NumPy

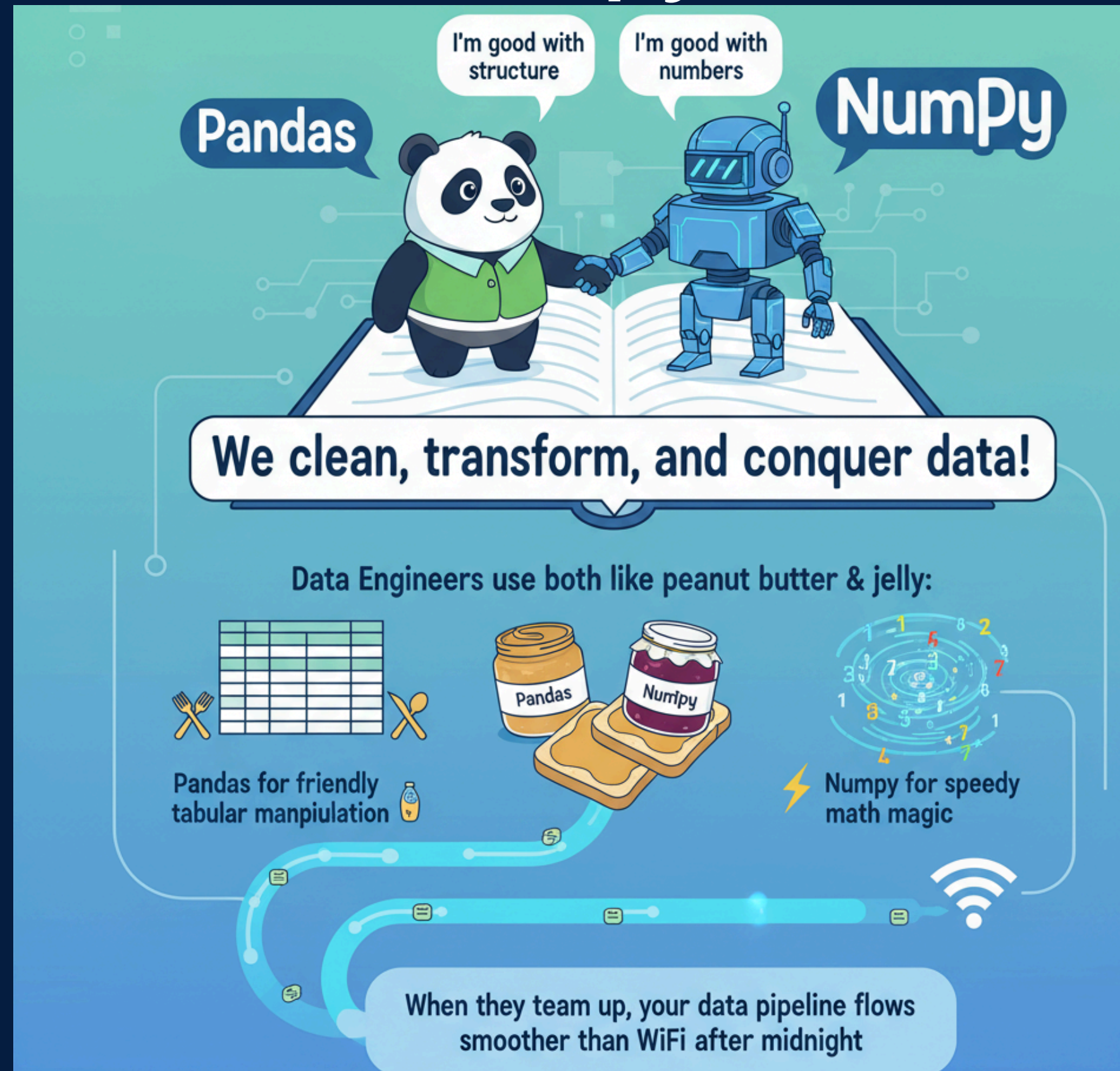




Common use cases of Numpy in Data Engineering:

- Efficient numerical computations on large datasets
- Handling large-scale data matrices or multi-dimensional arrays
- Optimizing performance in data transformation tasks
- Supporting advanced mathematical functions, such as matrix operations and linear algebra

When Pandas meets Numpy



**Thank You For
Your Attention !**